Haoyang Zhang

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Research Focus

I build efficient systems for AI on emerging hardware such as CPU–GPU superchips, co-designing LLM serving and training (Mixture-of-Experts inference, speculative decoding) with the memory hierarchy, interconnects, and GPU kernels. My background in computer architecture and memory/storage systems spans the stack, from devices and the OS up to serving frameworks.

Education

University of Illinois Urbana-Champaign 2022–2028 (expected)
Ph.D., Computer Science. Advisor: Prof. Minjia Zhang

University of Michigan, B.S.E., Computer Engineering (dual degree) 2020–2022

Shanghai Jiao Tong University, B.S.E., Electrical and Computer Engineering (dual degree) 2018–2022

Research Experience

ML systems for emerging hardware, co-primary author *2 papers under review; details on request*

Memory and storage systems, first or co-primary author *NeurIPS'25, HPCA'25, MICRO'23*

- **TeraIO:** Built lifetime-aware tensor offloading that extends GPU memory with SSDs via GPUDirect Storage for LLM training; 1.47× faster on average than ZeRO-Offload/Infinity (80.7% of ideal).
- **SkyByte:** Co-designed the host OS and SSD controller of a memory-semantic CXL-SSD; 6.11× faster than current CXL-SSDs with 23.08× less flash I/O traffic on average.
- **G10:** Built a unified GPU memory–storage architecture with compiler-scheduled tensor migrations; up to 1.75× faster than prior GPU memory solutions (90.3% of ideal).

Industry Experience

IBM Research, Yorktown Heights, NY May–Aug 2026
Research Intern, Hybrid Cloud. *Mentor: David Grove*

- Built an analytical cost model for IBM's Spyre dataflow AI accelerator, merged upstream into open-source torch-spyre, the PyTorch compiler backend, as a pass for comparing execution plans without running them (PR #3572).

T-Head Division, Alibaba Cloud, Hangzhou, China Jul–Aug 2021
Software R&D Intern (*Host: Yunhai Shang*). Optimized the JPEG library for RISC-V vector processors.

Skills

GPU & Kernels	CUDA (Graphs, Virtual Memory Management, TMA), CuTe DSL, Nsight / CUPTI
ML Systems	vLLM internals, FlashInfer, PyTorch, DeepSpeed; speculative decoding, MoE expert parallelism and offloading, KV-cache management
Memory & I/O	CXL, GPUDirect Storage, NVLink-C2C, InfiniBand, UVM / ATS, MPAM / resctrl
Architecture	NVIDIA GH200 / H200; gem5, Accel-Sim (GPGPU-Sim), MacSim, DRAMSim2
Languages	Python, C/C++, Verilog/SystemVerilog

Publications [G]

* co-primary authors

Under Review

2 papers under review (ML systems; details on request).

Conference and Journal Papers

[1] Ziqi Yuan*, **Haoyang Zhang***, Yirui Eric Zhou*, Apoorve Mohan, I-Hsin Chung, Seetharami Seelam, and Jian Huang. "**TeraIO: Cost-Efficient LLM Training with Lifetime-Aware Tensor Offloading via GPUDirect Storage**". In: *Conference on Neural Information Processing Systems* **(NeurIPS)**. 2025.

[2] **Haoyang Zhang***, Yuqi Xue*, Yirui Eric Zhou, Shaobo Li, and Jian Huang. "**SkyByte: Architecting an Efficient Memory-Semantic CXL-based SSD with OS and Hardware Co-design**". In: *IEEE International Symposium on High-Performance Computer Architecture* **(HPCA)**. 2025.

[3] **Haoyang Zhang***, Yirui Eric Zhou*, Yuqi Xue, Yiqi Liu, and Jian Huang. "**G10: Enabling An Efficient Unified GPU Memory and Storage Architecture with Smart Tensor Migrations**". In: *IEEE/ACM International Symposium on Microarchitecture* **(MICRO)**. 2023.

[4] Jiacheng Ma, Gefei Zuo, Kevin Loughlin, **Haoyang Zhang**, Andrew Quinn, and Baris Kasikci. "**Debugging in the Brave New World of Reconfigurable Hardware**". In: *ACM International Conference on Architectural Support for Programming Languages and Operating Systems* **(ASPLOS)**. 2022.

[5] Xingyue Qian, Jian Shi, Li Shi, **Haoyang Zhang**, Lijian Bian, and Weikang Qian. "**Scheduling Information-Guided Efficient High-Level Synthesis Design Space Exploration**". In: *IEEE International Conference on Computer Design* **(ICCD)**. 2022.

[6] Chen Chen, **Haoyang Zhang**, Kaicheng Guo, Xingzi Yu, Weidong Qiu, Zhengwei Qi, and Haibing Guan. "**Exploring Efficient Hardware Accelerator for Learning-Based Image Compression**". In: *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* **(TCAD)** 44.6 (2025).

Honors, Teaching & Service

MICRO 2023 Travel Grant 2023
Roger King Scholarship, University Honors, and Dean's List, University of Michigan 2020–2021
Graduate TA, Computer Architecture (CS 233), UIUC FA 2024, SP 2025, FA 2025
TA, Computer Architecture (VE 470) and Honors Physics (VP 160), SJTU SU 2022, SU 2020
Artifact Evaluation Committee, HPCA 2025